

Joshua Heinstein

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EDUCATION

Harvey Mudd College

Bachelor of Science in Engineering
Dean's List, GPA: 3.65

May 2026
Claremont, CA

Relevant Coursework: Digital Logic and Computer Architecture, System on Chip Design, Embedded Systems, Advanced Systems Engineering II, Electronic Circuits and Devices, Experimental Engineering

EXPERIENCE

Systems Engineer | Infinite Machine - New York, NY

July 2026

- Investigated and resolved critical vehicle field failures (random shutoffs) by deploying custom loggers, analyzing telemetry, and isolating a rogue CAN bus command triggered by a faulty mobile app update.
- Designed, manufactured, and deployed a fleet of 12 Teensy 4.1-based vehicle data loggers for an EV platform.
- Developed an accompanying desktop application featuring CAN decoding for real-time log analysis, charting, and raw-frame views.
- Iteratively designed and 3D-printed a vehicle-mount enclosure under strict packaging constraints, integrating custom cable strain relief.

Team Lead and Student Engineer | XDemics Corporation - Claremont, CA

Sept 2025 – May 2026

- Led a team of 6 to design a multi-axis robotic system to automate bioreactor cell seeding, growth, fluidizing, and harvesting.
- Achieved >95% recovery when testing with live cells by replacing a manual process with an automated 5-minute vibration protocol.
- Facilitated communication between student team, advisor, and XDemics' CTO to ensure satisfactory completion of project goals.
- Validated fluid dynamic performance using 2D COMSOL simulations and empirical testing, ensuring 100% surface coverage.

Systems Engineer | Amazon Prime Air - Claremont, CA

Jan 2025 – May 2025

- Implemented and tested ground and air based hardware/software to reduce drone GPS/GNSS reliance near tall buildings.
- Developed a Python-based computer vision system to target millimeter-level accuracy at 40 meters in drone 3D localization and heading determination.
- Tested Ultra-wideband based localization methods on drones, achieving cm-level accuracy with off-the-shelf drones.
- Produced a deployable solution and comprehensive 100-page report (trade-offs, KPIs, field data); handed off for review.

Structural Engineering Intern | Kimley Horn - West Palm Beach, FL

Jun 2024 – Aug 2024

- Designed over 10 structures for clients, ranging from foundations for industrial machinery, to retaining walls for a marina.
- Inspected the safety of over 30 buildings in Miami-Dade county and submitted detailed reports for each.
- Composed professional calculation packages and drafted CAD blueprints to be signed & sealed by licensed engineers.

Engineering Design and Manufacturing TA, Grader | Harvey Mudd College - Claremont, CA

Jan 2024 – May 2024

- Taught foundational teamwork skills to improve the process of creating a final product for a client.
- Provided individual feedback to over 50 students on multiple assignments per week.
- Composed a weekly memo summarizing crucial information that students often overlooked.

SKILLS

Equipment and Electronics: FPGA, MCU, STM32, ARM Cortex-M7, Embedded Systems, Oscilloscope, Soldering

Programming Languages: SystemVerilog, RISC-V Assembly, C, C++, Python, Java, R, MATLAB

Design Tools/Skills: Quartus, Questa, Git, Linux, RTL, CAN, LabVIEW, Simulink, KiCAD, Microsoft Suite, Autodesk, AI/LLM tools

Languages: English, Spanish (Fluent), German (Conversational)

PROJECTS

LED Zeppelin Audio Visualizer & Synthesizer

Nov 2025 – Dec 2025

- Engineered a dual-platform system using an STM32 MCU and iCE40UP5K FPGA to process and visualize real-time MP3 audio data.
- Implemented a 512-point FFT via the CMSIS-DSP library on the MCU to extract dominant frequencies, configuring the FPGA to map instantaneous frequency counts into 12 linearly-spaced bins to dynamically drive an RGB LED display via PWM.

RISC-V Multicycle Processor

Nov 2024 – Dec 2024

- Designed and implemented a fully-functional 32-bit RV32I compliant microcontroller using SystemVerilog.
- Created all components including the Arithmetic Logic Unit, Control Unit, Memory Interface and more.
- Leveraged simulation tools for testing and verification of the processor, ensuring RISC-V ISA compliance.

Autonomous Oceanic Surface Robot

Jan 2024 – May 2024

- Led a team of 4 in designing a GPS-equipped autonomous robot to measure wind speed and turbidity at preset locations in the ocean.
- Created analog circuits around sensors that were used in conjunction with a Teensy 4.0 MCU running custom C++ code.
- Used MATLAB, Excel and Python libraries to process and analyze data, write a technical report, and present our findings.